Object Detection of Glacial Features in ArcGIS

Opportunities in Glacier Investigation Program

Intern: Srinidhi Muralidharan

Overview

Glaciology is the study of glaciers, ice sheets, and other ice formations. Analyzing the movement and physical properties of glaciers can help researchers understand many natural phenomena. For example, glaciers are critical freshwater reservoirs, indicators of global warming, and carve valleys and cirques into earth's landscape. Through glacial features like gullies, alcoves, and cirques, we can learn how landforms evolve.

Gullies are characterized as small channels carved by running water or water erosion. More developed gullies imply longer deglaciation periods while fewer or less developed gullies indicate a recent glaciation. Moreover, gullies can estimate how long an area has been ice free. Similarly, alcoves are hollowed out areas that are fan shaped found near sides of slopes and valley walls. Alcoves form through weathering, erosion, and debris flow. Alcoves are crucial for the formation of gullies since they are the starting point or source area for gullies. Likewise, cirques are bowl shaped depressions at the heads of mountain valleys. These bowls without any ice can help researchers predict the elevation of a glacier where the accumulation of snow balances the loss of snow, also known as the equilibrium line altitude or ELA.

In turn, alcoves, gullies, and cirques aid researchers in assessing how glaciers have evolved over time and have responded to climate change. We can specifically look at the Trollaskagi Peninsula in Iceland to analyze these features and even compare them to terrain in Mars.

The Trollaskagi or Troll Peninsula in northern Iceland is a region with similar geographic features as to Mars. The Troll Peninsula and the Deuteronilus Mensae region in Mars share basaltic plateaus. These plateaus are flat areas of volcanic rock with many gullies, cirques, and rock glaciers. The current understanding of glacial features on Mars is that they are cold-based or the ice stays frozen and has minimal erosion. However, the alcoves on Mars closely resemble Trollaskagi's cirques which are known to be formed from warm-based or polythermal glacial erosion. This similarity raises the question of whether Martian glaciers experienced warm-based glacial erosion causing cirque-like alcoves. Furthermore, the proportion of cirques, alcoves, and gullies is important, as their formation progresses from gullies to alcoves to cirques. Gullies form into alcoves significantly slower than alcoves to cirques. Knowing this, it might be useful to see the proportion of these glacial features in the Troll Peninsula.

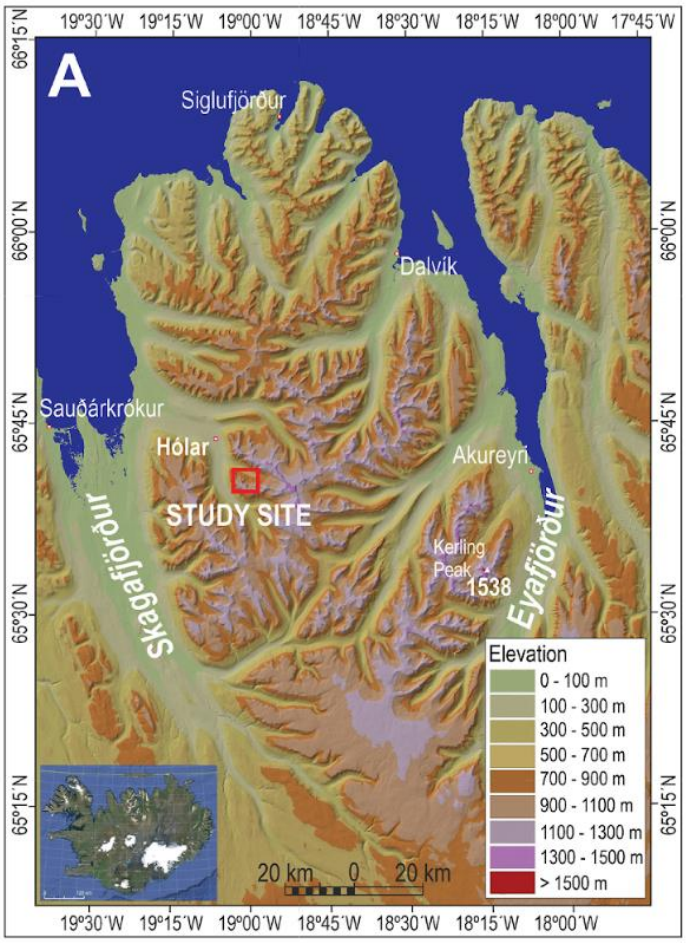


Figure I: Trollaskagi Peninsula

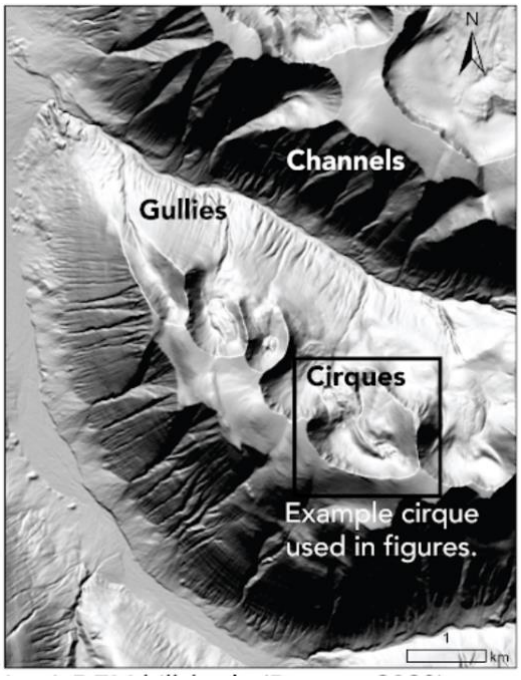


Figure II: Hill shade of Glacial Features

ArcGIS Mapping

My project primarily uses object detection on the Troll peninsula to automate identification and mapping of gullies, alcoves, and cirques. This accelerates the process of mapping glacial features which is a tedious process. By training models in the Troll peninsula, researchers can apply the same methods to Mars and continue to identify and analyze the similarities between Earth's and Mars' glacial features. I began by creating feature classes in ArcGIS for cirques, alcoves, and gullies, which I then mapped using the polygon tool. I had mapped about 300 to 400 gullies and cirques as well as 170 alcoves. Specific patterns that I could see was there were an abundance of gullies and cirques in the Trollaskagi peninsula while there weren't many alcoves. However, whenever there were alcoves, there were batches of alcoves in an area or none.

Using these feature classes as training data, I used three specific geoprocessing tools in ArcGIS to train the deep learning model to detect gullies, alcoves, and cirques. First, I used the "Export Training Data for Deep Learning" tool to export the data into image chips which are small

patches cropped from the main imagery, label files that store information identifying each feature, and a metadata file with dataset structure and label classes. Once this folder is created, I used the “Train Deep Learning Model” tool. This tool trains a neural network using image chips, labels, and metadata from the exported data. I selected the object detection model Mask R-CNN as the model type which detects.

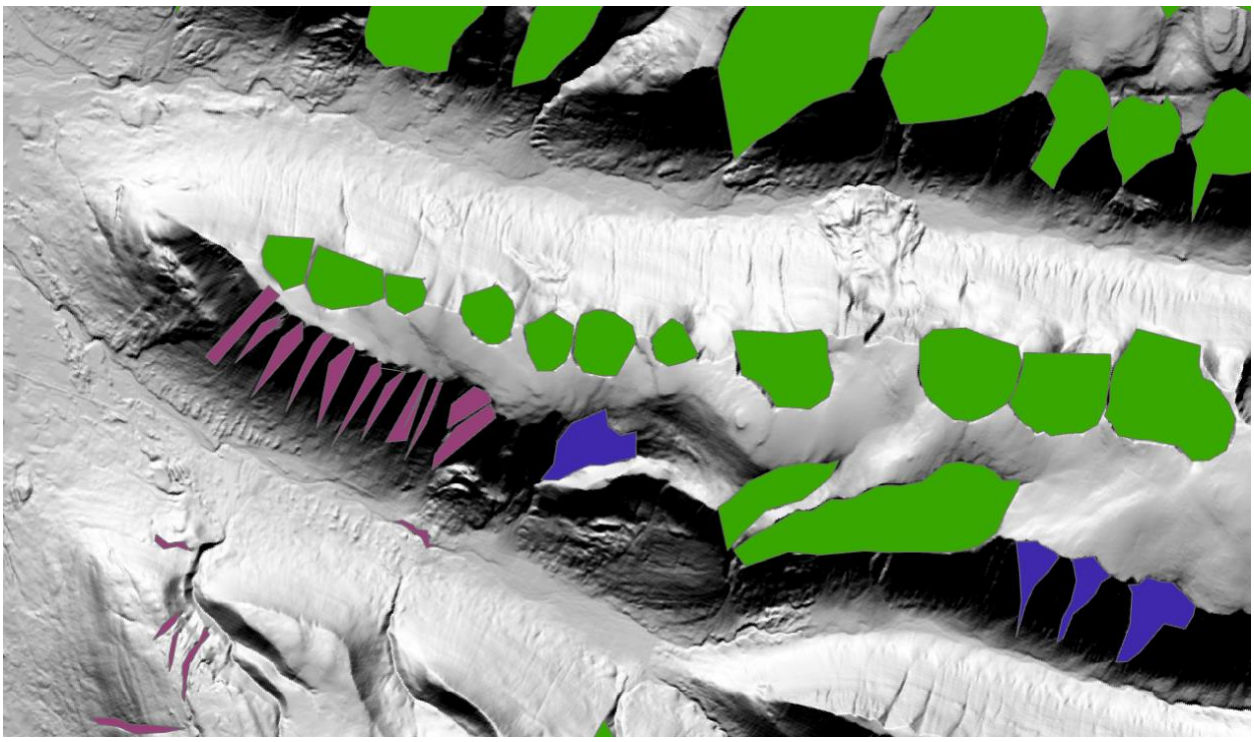


Figure III: ArcGIS Feature Classes of Gullies, Alcoves, and Cirques

Mask R-CNN Model

I decided to use the Mask R-CNN which stands for Mask Region-based Convolutional Neural Network. This is a type of deep learning model used for object detection in computer vision tasks. Mask refers to pixel-wise segmentation where instead of the object being boxed, it is masked on the pixel level. The R refers to a region or part of the image that is likely to contain the object. CNN or Convolutional Neural Network takes an image and outputs probabilities on possible objects. If the object that we chose is in the image, the CNN gives back a high probability. The advantage of using deep learning with CNNs over machine learning is that there is no need to manually extract features through code. Instead, with CNN, it’s fed raw images, and CNN learns the best features through training. Through instance segmentation, CNN

recognizes objects within an image and outlines each object at the pixel level instead of drawing bounding boxes. (Figure 3)

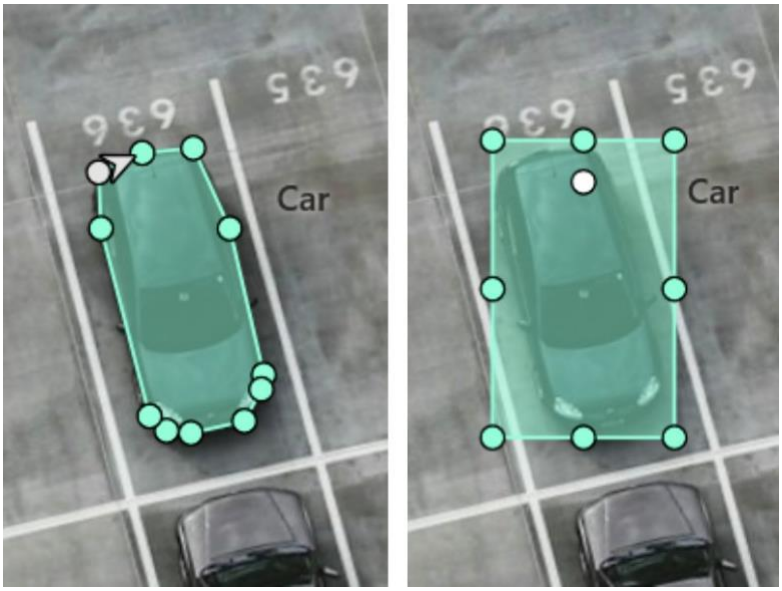


Figure IV: Polygon Annotation versus Bounding Box

Mask R-CNN uses a neural network which in summary has an input layer, hidden layers, and an output layer. In the input layer, data goes in or in our case, an image. The model then scans the image and proposes regions where a glacial feature might be. Next, in the output layer, the model decides if each region has a glacial feature or if it's just the background. Lastly, instead of drawing a box, the Mask R-CNN model draws the exact shape of the glacial feature and stores it as a prediction. The hidden layers in Mask R-CNN will detect basic patterns, then shapes, and glacial features' parts. Each layer is the input for the next layer as it finally reaches the final decision in predictions.

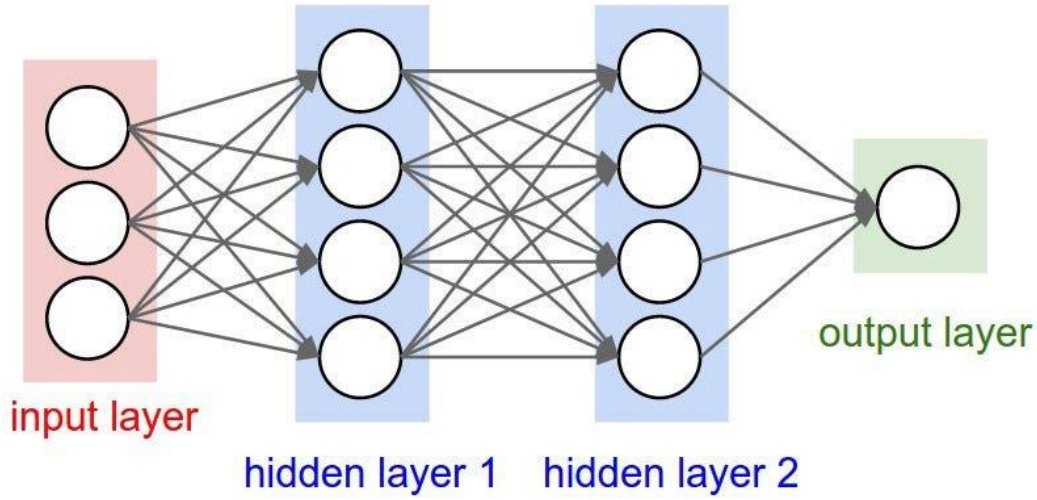


Figure V: Neural Networks' Layers

In Mask R-CNN's architecture, it builds on Faster R-CNN but has its own features. The main difference with Faster R-CNN is that it detects an object and draws a bounding box over it, while Mask R-CNN detects an object and draws the exact shape of the object on a pixel level. (Figure III). Moreover, when a model detects Regions of Interest (ROIs), each region of interest has different sizes. However, in the hidden layers, a fixed size will be used from the images and for the model to be trained. ROI Pooling is used in Faster R-CNN which takes the ROI's coordinates from the image, divides into a fixed grid 7x7, and applies max pooling (In ROI, max pooling converts the proposed region into a fixed size by dividing the cells and keeping the maximum value from each cell's pixels). This causes rounding of coordinates or a quantization error. Quantization error is when small shifts in the location of the object in the image can alter the object's shape when training the model. Instead of ROI Pooling, Mask R-CNN uses ROI Align where it removes quantization error by using floating point values rather than rounding.

Mask R-CNN Loss Functions

Similarly, Mask R-CNN trains with loss functions that improve the model. In Mask R-CNN, classification loss is used to tell the model if the proposed region has the correct object. Next, bounding box loss is used to see how well the object and the predicted box line up through the predicted bounding box and actual bounding box's coordinates (x, y, w, h). Lastly, the mask loss helps the model draw the correct mask of the object on the pixel level. This is through the image's pixels where each pixel has a probability from 0 to 1. These three functions focus on different aspects and combined; help improve the predictions of the model. Then, using the score

that the loss functions output, we can use the sum of the three functions to see the how well the model predicts. Simply put, the equation is the sum of the three functions with weights on each part. A lower total loss means the model's predictions are closer to the true value.

$$Total\ Loss\ =\ Classification\ Loss\ +\ Bounding\ Box\ Regression\ Loss\ +\ Mask\ Loss$$

$$Total\ Loss\ = w_1\ L_{cl}\ +\ w_2 L_{bbox}\ +\ w_3 L_{mask}$$

Figure VI: Loss Equations

In ArcGIS, these loss values are combined in the training report as training loss and validation loss for per epoch. Training loss refers to how well the model fits the training data during each epoch. The lower the value, the model's predictions are closer to the actual truth. Next, the validation loss measures performance on a separate dataset that the model hasn't been trained on. If the validation loss is much higher than the training loss value, this can indicate overfitting which is when the model memorizes training data rather than learning patterns. Lastly, the training report outputs average precision. This is the accuracy of the model in terms of how the predicted masks match the actual truth. The higher the value, the better for the model.

ArcGIS Results

I mapped 349 cirques, 170 alcoves, and 401 gullies. This is a fairly small sample size for training data. However, I wanted to explore how we can utilize deep learning to ease the mapping process with a small area already mapped. In ArcGIS, the first round of testing brought us average results. The alcoves that the deep learning model had mapped were extremely small compared to the actual size of the alcove while there were only a handful that the model had mapped instead of hundreds. I ran the model on each glacial feature separately and found that the model worked best for cirques comparatively. This is most probably due to the size and uniform appearance of cirques. Compared to alcoves and gullies, cirques have a repetitive look while alcoves and gullies form in a variety of shapes and sizes. The model may require more training data to learn these distinctions for glacial features.

In ArcGIS, the model's results are presented through training loss and validation loss. When we examine the training loss per epoch, it's apparent that there's a large decline from 0 to 1 while the rest of the epochs have a gradual decline. When reviewing the validation loss, it's apparent that there's a gradual decline as well but with a shallower slope.

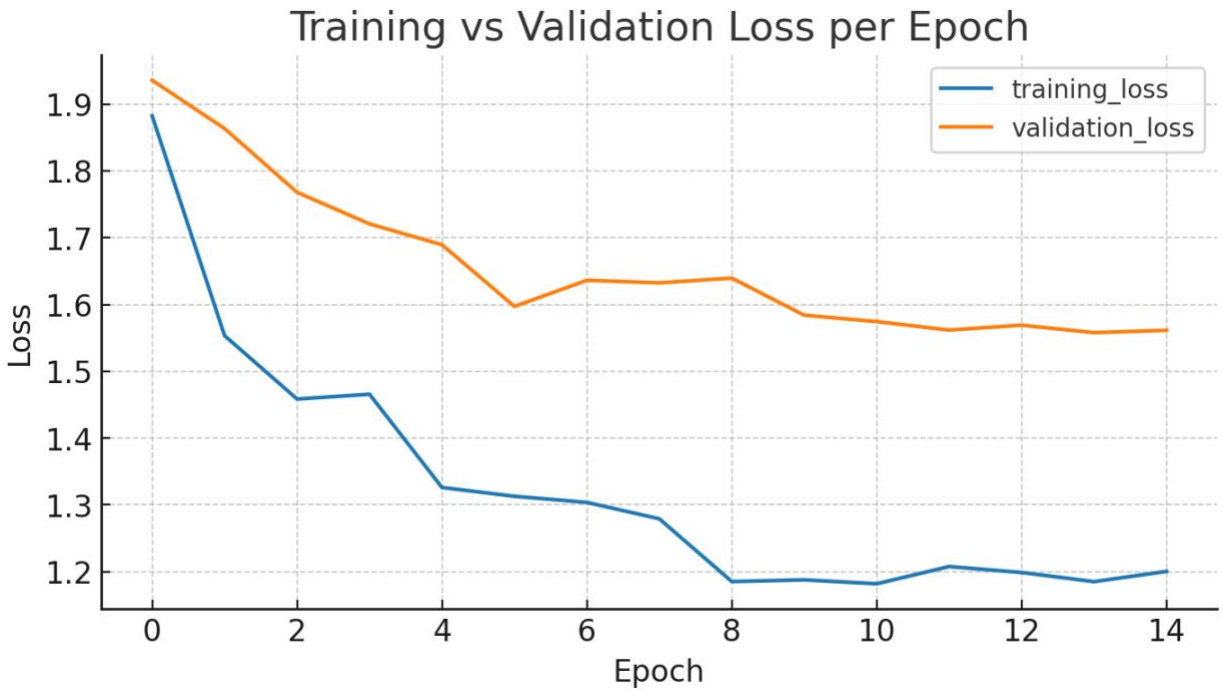


Figure VI: ArcGIS Training vs Validation Loss

The gradual declines in both the training and validation loss refer to the model's predictions getting closer to the actual truth after each epoch and that the model is learning patterns from the training data and not memorizing it. Validation loss is decreasing at a slower rate than the training loss which means that the model is improving on unseen data while not generalizing the predictions. Also, the two losses stay together in an ideal learning curve which means that learning increases quickly early on and then slows down to eventually reach a plaetueau. This is important because this is how the model will learn without generalization.

In addition, average precision is a metric used to measure the precision-recall curve into a single number. Precision tells us, out of all the model's predictions, how many predictions are truly accurate. Recall tells us, out of all of the actual glacial features present, how many the model successfully predicted. Average precision balances both precision and recall by using confidence scores from each prediction to check performance through different confidence score cutoffs. The average precision is a single score that summarizes this into one overall measure.

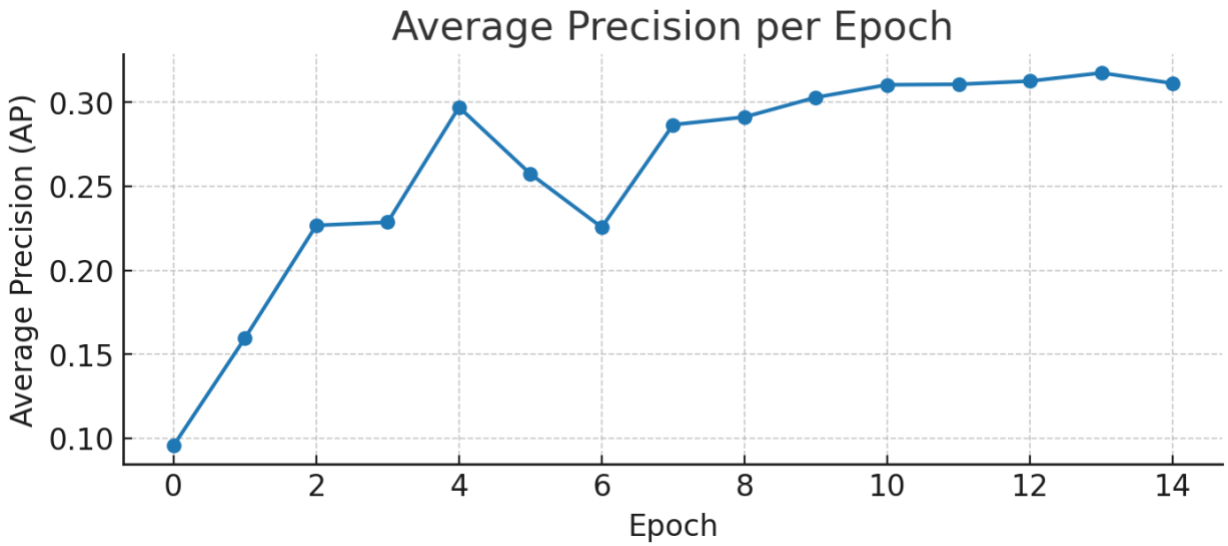


Figure VII: ArcGIS Average Precision

Through ArcGIS, the average precision reached 0.3 at the end of epoch 14. While the average precision can depend on the dataset, 0.7-0.9 indicates an accurate model while below 0.5 indicates that the model is missing true positives or making false predictions. To improve the model, I shifted from ArcGIS to using Detectron2. ArcGIS has tools that are simple to use and produces object detection results for glacial features. However, the process is extremely slow. The model can be fine-tuned more effectively through Detectron2 than in ArcGIS. For this reason, I pivoted to Detectron2 and used UW's supercomputer to run my Python scripts.

Detectron2 Model

Using UW's Hyak, I used Facebook's Detectron2 to train on my gullies data. Detectron2 is a computer vision library or toolbox that's written with PyTorch and includes Mask R-CNN implementation. Using Detectron2, I accessed Mask R-CNN to train the model on gullies. Some methods to fine-tune the model that I learned about are changing the learning rate, adjusting anchors, increasing mask resolution, and improving training data.

Learning Rate

To make sure the Python scripts run properly on Hyak, I only ran 2,000 iterations. However, training should be longer for better results. In the script, max iterations, steps, base_lr, and step size are important factors that enhance the model's prediction.

Firstly, when we change the max iterations from 2,000 to a larger number like 90,000, training time increases, which can improve predictions. Next, Base_lr is the starting learning rate (LR) or step size. With High LR, the model's accuracy may improve quickly in the beginning, like taking big steps when descending a mountain. However, the model might overshoot. The lower LR improves the model slowly, training is more stable, like taking smaller steps will take more time while descending, but might be safer. We are changing the starting learning rate from 0.00025 to 0.0005 so that there are slightly larger steps so that it doesn't take too many iterations to converge.

Regarding some base or default numbers, the base learning rate is usually set to 0.00025 in Detectron2. The max iterations' base is also 90,000 and for larger datasets, you can even input 120,000 plus.

Cfg.SOLVER.STEPS will change the step size during training. Early on, big steps help the model learn. However, as time goes on, you need smaller steps to carefully fine-tune. We've set it to 60,000 to 80,000 which means from 0-59,999 iterations, the learning rate is 0.0005, from 60,000 to 79,999, it's 0.00005, and over 80,000 iterations, the learning rate is 0.000005. The learning rate is getting smaller after iterations so that the model ends up with more precision.

Adjust Anchors

Anchors are predefined boxes used to search for objects in the image. The network places multiple anchors of different sizes and predicts if there's a gully there, for example. However, default anchors in our model are square shaped and fail to miss small gullies. The code needs to be adjusted so that gullies that are small, thin, or oddly shaped can be identified. We can change the anchor generator size and aspect ratios so that extremely small and large glacial features can be identified. The anchor size is about the general scale of the box, like 8x8 or 16x16 pixel boxes, while aspect ratio is the width to height ratio of the box.

Increase Mask Resolution

Mask-RCNN predicts a low-resolution mask, 14x14 pixels by default. Since gullies are very small features that are only a few pixels wide, the model can miss them. We can sharpen the predictions by increasing resolution. Instead of keeping 14x14, we can double the resolution to 28x28 to find smaller features.

Conclusion

In this summer project, I worked on combining geospatial analysis and object detection. Using satellite imagery and digital elevation models (DEMs), I mapped cirques, alcoves, and gullies in ArcGIS, which I then exported as training data. I implemented ArcGIS's in-house machine learning tools to test object detection with my training dataset. Later, I moved to Detectron2 (Mask R-CNN) in Python to fine-tune and improve that model. To optimize performance, I trained the model on UW's Hyak supercomputer, while leveraging AI tools to accelerate development. Finally, I documented the process in a technical guide with the detailed pipeline and strategies for refining the object detection model for various glacial features and terrains.

Sources

Li, A., Koutnik, M., Brough, S., Spagnolo, M., and Barr, I.: Cirque-like alcoves in the northern mid-latitudes of Mars as evidence of glacial erosion, EGUsphere [preprint], https://doi.org/10.5194/egusphere-2023-2568, 2024.

Figure 4, *Polygon annotation vs. bounding box example.* Reprinted from *Accurate object detection with Polygon annotation*, by V. Kapoor, 2024, *Mindkosh* (https://mindkosh.com/blog/accurate-object-detection-with-polygon-annotation/). Copyright 2024 by Mindkosh.

Tanner, G. (2020, May 11). *Getting started with Mask R-CNN in Keras* [Blog post]. Retrieved from https://gilberttanner.com/blog/getting-started-with-mask-rcnn-in-keras/

Insane Calling. (n.d.). *Day 5 – Tröllaskagi Peninsula*. In *Iceland itinerary*. Retrieved August 11, 2025, from https://www.insanecalling.com/europe/iceland/itinerary/day-05/